KRIYA ZERO

The One-Demonstration Physical Skill Compiler

Smart Education • Team XYRO • iQOO Pune 2026

Show it once. Learn it anywhere. Prove you can do it.

Problem

Practical learning is difficult to scale because tutorials can demonstrate a procedure but cannot prove that a learner reproduced it correctly. Labs, ITIs and vocational programs still depend heavily on instructor supervision and manual practical assessment.

What we are building

KRIYA ZERO is an Android-first, phone-native system that converts a single expert demonstration of a structured physical task into an executable Skill Capsule. The phone observes the workspace, captures learned checkpoints, compiles ordered dependencies, then watches another learner perform the procedure and detects wrong, skipped or out-of-order states. The same representation powers guidance, verification and an evidence-backed skill report.

Core innovation

Demonstration	→	Executable procedural graph	→	Live verifier
Expert performs once		States + steps + dependencies + checkpoints		PASS / FAIL / sequence error / correction

Why it is different

Conventional videos teach. Digital SOP and AR work-instruction tools guide procedures that are authored beforehand. KRIYA's wedge is zero-authoring physical verification: a fresh demonstration creates the verifier itself. The hackathon claim is intentionally narrow and defensible — visually observable, structured tabletop procedures rather than arbitrary open-world human activity.

Technical approach

CameraX provides live phone-camera input. The current prototype learns compact visual checkpoint fingerprints with five-frame stabilization, compiles them into a Skill Capsule, and uses an explicit deterministic Verification Engine for PASS/FAIL and temporal sequence checks. AI/perception supplies evidence; it does not unilaterally decide competence. This makes local embedding or narration models replaceable without rewriting assessment semantics.

Current prototype status

Implemented: Android + Jetpack Compose shell, live rear-camera flow, Teach → Compile → Verify → Assessment journey, learned visual checkpoints, brightness-normalized similarity, temporal skip detection, correction without restart, evidence-backed metrics, local persistence backend, unit tests, GitHub Actions and CI-generated debug APK. Real-device threshold calibration on the hackathon iQOO phone remains a required validation step.

KRIYA ZERO — Phase 1 Execution

20-second proof

1) Expert demonstrates a 3–5 state tabletop task. 2) KRIYA creates checkpoints. 3) Workspace resets. 4) Learner performs Step 1 correctly. 5) Learner intentionally makes or skips Step 2. 6) KRIYA rejects the state. 7) Learner corrects it without restarting. 8) KRIYA verifies completion and generates the skill report.

Who benefits

Engineering labs	Circuit assembly, instrumentation and practical exercises
ITIs / vocational programs	Scalable repetition and standardized practical assessment
Industrial workforce	Onboarding and procedure practice where expert time is scarce
Field technicians	Guided, measurable procedures without constant remote supervision

Why iQOO

The phone is the product: camera sensing, local perception, offline-critical execution and real-time feedback are central rather than decorative. The hackathon build can use the current deterministic visual layer as the safety net while strengthening perception with a local image-embedding model and local narration on the iQOO device.

Feasibility & scope

The MVP deliberately focuses on structured tabletop procedures with visually distinguishable state changes. This keeps the 30-hour build testable and demo-safe while preserving a scalable architecture for richer local models, multi-view verification, multilingual guidance and Skill Capsule sharing later.

Safe novelty statement

KRIYA ZERO learns previously unseen, visually observable, structured physical procedures from a single demonstration and automatically creates reusable guidance and verification checkpoints without manual procedure authoring or task-specific pass/fail training.

Repository

https://github.com/omghotekar01-dotcom/KRIYA-ZERO

Prototype code, tests, architecture, demo script, on-device acceptance protocol and CI/APK workflow are included in the repository.